The current issue and full text archive of this journal is available on Emerald Insight at:
https://www.emerald.com/insight/1066-2243.htm

Received 4 November 2019
Revised 12 May 2020
13 September 2020
19 December 2020
Accepted 19 December 2020

Understanding automated conversational agent as a decision aid: matching agent's conversation with customer's shopping task

Jengchung Victor Chen, Huyen Thi Le and Sinh Thi Thu Tran
Institute of International Management, National Cheng Kung University, Tainan, Taiwan

Abstract

Purpose – To provide better services to customers, especially immediate responses and 24/7 availability, businesses are implementing text-based automated conversational agents, i.e. chatbots on their social platforms and websites. Chatbots are required to not only provide customers with necessary consultancy and guidance but also communicate friendly and socially. Based on the cognitive fit theory, this study attempts to examine the role of chatbot as a decision aid and how the match between information presentation in forms of decisional guidance and communication style and the shopping task influences consumers' perceived cognitive fit and decision performance outcomes.

Design/methodology/approach – A 2 x 2 x 2 between subject online experiment was conducted to identify which kind of decisional guidance (suggestive and informative guidance) and communication style (task-oriented vs social-oriented style) are the most appropriate for each type of shopping task (searching vs browsing task).

Findings – The findings show that when customers interact with chatbots, they will perceive higher cognitive fit if the chatbots provide them with suggestive guidance and communicate in a friendly style especially when they perform a searching task.

Originality/value – This study is the first attempt to understand the role of chatbots as a decision aid to customers using the communicative language. This study also tries to explore the cognitive fit theory in a novel way, and we propose the information presentation in forms of communicative language rather than matrices, tables and graphs.

Keywords Decisional guidance, Communication style, Shopping task, Cognitive fit, Decision outcomes, Chatbot

Paper type Research paper

1. Introduction

Together with product attributes, the customer service experience is becoming more and more important. HubSpot's survey in 2018 found that 82% of consumers wanted immediate response for their marketing or sales questions and 90% for customer service questions (Hubspot, 2018). According to Microsoft's survey in 2013, a seller had only about 8 seconds to get the attention of a customer, compared to 12 seconds in 2000. And that seller would lose about 50% or more of his potential customers with any delay. Consumers are becoming impatient and sellers need 24/7 customer supports. Virtual agents have emerged as a possible solution that could employ a remarkable effect in this situation. According to research of Business Insider Intelligence, US Office of Personnel Management, using virtual agents is expected to help the company save 36% sales representative expense and 29% customer service representative expense. The involvement of the virtual agents in providing product information and guiding customers through the purchase process could reduce the impersonal feeling of online purchases (Verhagen *et al.*, 2014).

As social commerce is gaining popularity, many companies have started offering services through social media fanpage and messenger to their customers. According to Twilio, about 66% of consumers want to be contacted via messaging applications. Text-based automated

Internet Research
Vol. 31 No. 4, 2021
pp. 1376-1404
© Emerald Publishing Limited
1066-2243
DOI 10.1108/INTR-11-2019-0447

conversational agents (ACAs) or chatbots are commonly implemented on social media, such as Facebook, Twitter, etc. and commercial websites (Araujo, 2018). Moreover, 24/7 service chatbots provide a convenient way for customers to reach out to companies and for companies to provide better services (Brandtzaeg and Følstad, 2018; Feine *et al.*, 2019b; Van den Broeck *et al.*, 2019). Although chatbots have emerged for decades, the real buzz of this technology started in 2016 when it was integrated into the mobile messaging applications of some social networking sites, for example, Facebook Messenger, Telegram, Slack, Kik and Viber. ACAs are designed to communicate with humans using natural language (Dale, 2016; Feine *et al.*, 2019a; McTear *et al.*, 2016), so language for communication is crucial to ACA designs. ACAs are required to not only provide customers with necessary consultancy and guidance but also communicate friendly and socially (Feine *et al.*, 2019b; Go and Sundar, 2019).

The efforts of researchers on ACAs to mimic human agents make human–chatbot conversations flow more and more naturally and share similarities with human–human conversations, in terms of emotional, relational, psychological outcomes (Ho *et al.*, 2018), the communication quality (Edwards *et al.*, 2014) and customer satisfaction (Verhagen *et al.*, 2014). Chatbots are replacing human agents in taking care of customers and researchers and practitioners are putting efforts into mimicking human's natural language and applying in ACAs, generating the most similarity among ACAs and human agents. However, previous studies also show that when interacting with an ACA, users and customers behave differently compared to when their partner is a human agent, such as using shorter sentences (Hill *et al.*, 2015), losing interest (Fryer *et al.*, 2017) or even purchasing less (Luo *et al.*, 2019). Since users and customers could easily detect if they are conversating with a virtual agent, instead of generalizing the results of human–human interaction into human–chatbot interaction, it is necessary to study the chatbot designs and investigate the consumers' experience of interacting with the ACA to understand consumers' favorable behaviors.

Despite the findings of previous research regarding the effects of ACA appearance on commercial websites (Gulz and Haake, 2006; Holzwarth *et al.*, 2006; Qiu and Benbasat, 2009; Wang *et al.*, 2007; Wood *et al.*, 2005; Zhu *et al.*, 2014) or ACA adoptions (Kumar and Benbasat, 2006; Qiu and Benbasat, 2009; Zhu *et al.*, 2014), lack of studies looks into the language designs of the text-based ACAs, i.e. chatbots, and how these designs help the customers in their decision-making process. Unlike other forms of ACAs, chatbots implemented on social media platforms are text-based, making verbal language the only means for chatbots to communicate with customers. Therefore, this study focuses on investigating the language designs of text-based chatbots in terms of decisional guidance and communication style and how these designs help to improve the consumer's decision-making performance in the context of online shopping. To be specific, we try to answer the research question: *For each type of shopping task (searching and browsing), which kind of decisional guidance and communication style are most accurate for ACA to help customers get the best decisional outcome?*

The purpose of this research is to improve the customer's decision performance with the support of ACAs. To do that, we start from the shopping behaviors of customers, which are represented by the searching and browsing task. We assume that when individuals perform each of the shopping tasks, they need the respective information representation provided by the assistant to complete such tasks. Decisional guidance and communication style are the most accurate information formats in the context of ACAs. Follow the cognitive fit theory, this matching of the information representation and the shopping task will lead to a cognitive fit, and this mental representation in cognition, in turn, will improve the decisional outcomes by increasing the decision quality and decreasing the decision effort.

Besides, to have better understandings of customers' decisional performance, especially potential influencers of the decision-making process, we include perceived task complexity as

the moderator. Perceived task complexity was found to have an impact on users' behaviors and was an important determinant to affect the effectiveness of various service technologies (Gupta *et al.*, 2013). Complicated tasks would discourage the consumers and reduce customer retention (Kamis *et al.*, 2008), negatively impact the performance and the enjoyment of customers (Xu, 2016; Xu *et al.*, 2014b). Different levels of task complexity would affect the psychology and behaviors of consumers differently during the decision-making process. Thus, this paper also attempts to investigate the impact of perceived task complexity as a moderator on how ACAs help consumers make decisions.

Our research contributes in several ways. To the best of our knowledge, this is the first study that focuses on the language design of the chatbots with a specific online shopping context. This study not only reveals ACAs' role as a decisional aid but also sheds light on the mechanisms to improve the decisional outcomes. Social-oriented communication style is found to be dominated in most cases, which means that customers mostly prefer to interact with a friendly and jolly chatbot. More specifically, if customers are in search of a specific product, suggestive and informative ways of communication both create the same effect if the chatbots use social-oriented communication style. However, in case customers are unsure and browse to find suitable products, the chatbots should be designed to friendly suggest specific products or use the task-oriented style to give all the needed information for them to choose. These findings are important to practitioners regarding the design of the language styles for their chatbots. Moreover, our novel way of conceptualizing the information presentation as decisional guidance and communication style and our introduction of the perceived cognitive fit construct extend further the understandings of the cognitive fit theory as well as the ACA literature.

2. Theoretical background and literature review

2.1 Automated conversational agents

The first conversational agent (CA) was ELIZA, a text-based agent developed by Joseph Weizenbaum as a computer program that "makes natural language conversation with a computer possible" (Weizenbaum, 1966, p. 36). Modern technology has developed the text-based CAs into voice-based dialog systems, voice user interfaces, embodied CAs (ECAs) and social robots (McTear *et al.*, 2016). Feine *et al.* (2019a) categorized CAs into text-based CA, voice-based CA and ECA. Text-based CAs are normally referred to as chatbots and implemented on commercial websites as well as the messenger function of social media platforms such as Facebook. Voice-based CAs are developed to be personal assistants and implemented on personal devices such as Apple's Siri, Microsoft's Cortana, Google's Assistant and independent smart speakers such as Amazon's Alexa (Dale, 2016). Different from text-based and voice-based CAs, ECAs imitate both human verbal and nonverbal communication and normally have a visual presentation such as a 3D animated avatar (Chattaraman *et al.*, 2019).

The implementation of ACAs brings several benefits to businesses and more and more businesses use ACAs as a marketing tool to improve their sales and customer relationship management. Verhagen *et al.* (2014) suggested that using virtual agents to give consumers product information and guide them through the purchase process could reduce online purchases' impersonal feeling. Van den Broeck *et al.* (2019) found that if consumers found the chatbot helpful and useful, they would feel satisfied with the conversations and less intrusive with the advertising messages sent by the chatbot. Several research studies demonstrated that the ACA appearance on a website could positively affect customer trust and satisfaction (Chung *et al.*, 2020), improve customer relationship (Beldad *et al.*, 2016; Qiu and Benbasat, 2009; Saad and Abida, 2016) and increase the productivity (Mimoun *et al.*, 2017). Poncin *et al.* (2013) in turn indicated that in the context of the 3D website, using virtual sales agents to

interact with customers improves consumer productiveness during the online shopping process, in form of effectiveness and efficiency. To improve the benefits of ACAs, it is necessary to make the ACAs interact in the same way as humans do. Go and Sundar (2019) suggested three ways to improve the humanness of ACAs, including using the human figures, giving ACAs an identity and mimicking human languages. For chatbots, the language mimic is even more important because verbal language is the only tool of communication.

2.2 Human-agents vs ACAs

Research in human–computer interactions shows that human–chatbot conversations share many similarities with human–human conversations. For example, when users disclosed their emotions in a conversation, the effects on emotional, relational and psychological outcomes were the same, regardless of whom, humans or chatbots, they talked to (Ho *et al.*, 2018). Edwards *et al.* (2014) compared the communication quality between human agents and bot agents on Twitter and found that Twitterbots showed to have the same source credibility, communication competence, interactivity as human agents but were less attractive. Previous studies continuously attempted to identify ACA's design cues such as visual cues (appearances or avatars), identity cues (gender, names or identity) and conversational cues (natural languages) to best mimic humans (Go and Sundar, 2019; Holtgraves *et al.*, 2007). Despite the efforts to mimic human behaviors and use chatbots to replace humans in some tasks, studies from early on have found that users' behaviors differ when they interact with chatbots, comparing to when they interact with a human. In terms of trust-building, even though users had the same trust for virtual agents and human agents, the processes of building trust were significantly different (Komiak *et al.*, 2004). Hill *et al.* (2015) compared 100 instant messaging between users and human agents with 100 messages between users and Cleverbot and found that human–chatbot conversations tended to be shorter in terms of the number of words per sentence and number of sentences in the whole conversation with less vocabulary richness. Fryer *et al.* (2017) conducted an experiment where students were paired up with studying partners, either humans or chatbots, for 12 weeks. At the end of every 3 weeks, the students' interest was measured. The results showed that the interest stayed the same for students with human partners while significantly dropped for students with chatbot partners. In the e-commerce context, well-trained chatbots could be as effective as proficient service workers, but customers preferred to interact with human agents and purchase rate reduced by 79.7% if customers knew they were interacting with a chatbot (Luo *et al.*, 2019).

Based on the comparisons above, we argue that users and consumers react differently to human agents and virtual agents, and it is necessary to investigate their behaviors toward ACAs besides applying related results about their behaviors toward human agents. Because the goal of ACA development is to best mimic human conversations, researchers must borrow human skills and characteristics and apply them to ACA designs. However, even if the imitation of such human skills and characteristics is well-deployed, users and customers can detect whether they are interacting with a human or virtual agent, and, as a result, behave differently. Moreover, unlike human agents who could react flexibly to situations, ACAs, especially low AI text-based chatbots, could only react as programmed. Therefore, to have a better understanding of not only which types of designs are more effective but also how users and customers behave with the ACA supports, researchers essentially test on the ACAs themselves to curb the variation of generalizing the results that are found by using human agents.

2.3 Theoretical background

2.3.1 Cognitive fit theory. The cognitive fit theory proposes that a good match between the problem-solving task and problem presentation creates a higher problem-solving

performance (Vessey, 1991). According to the theory, the task refers to a problem that the individual needs to solve, while problem representation is the way the information is presented as an aid to solving that task (Chan *et al.*, 2012). The cognitive fit theory has been studied over the years in a variety of contexts, and the information presentation (problem representation) was in form of tables or graphs (Hirsch *et al.*, 2015; Kumar and Benbasat, 2004; Mertins and White, 2016; Vessey and Galletta, 1991), matrix versus list (Hong *et al.*, 2004), geographical maps (Khatri *et al.*, 2006), pictures (Perdana *et al.*, 2018) and audiovisual formats (Lee *et al.*, 2001) or multimedia presentation (Jiang and Benbasat, 2007). This study expands the information format of the cognitive fit theory to decisional guidance and communication style, which fit with the scope of the ACA context.

The correspondence between task and information representation is considered as the mental representation or the fit cognition that mediates the performance of the individual (Chan *et al.*, 2012; Shaft and Vessey, 2006; Vessey, 1991). The mental representation can be considered as a process or an output perspective. Under the process perspective, mental representation refers to the process that people use in cognition during understanding and solving the problem process (Shaft and Vessey, 2006). Mental representation is the cognition inherent to problem-solving and exists even if the problem representation and task do not fit (Chan *et al.*, 2012). The cognitive fit theory states that when the problem representation accentuates the same types of problem-solving information and processes with the task, it creates a cognitive fit (Chan *et al.*, 2017) then a consistent mental representation is produced and leads to more accurate and effective decision-making performance (Hong *et al.*, 2004). If the information presentation format and the task do not match, people must try to convert the information into a more useable format. This process might require a greater effort and lead to a decrease in decision performance, in terms of decision accuracy and/or decision-making time (Kelton *et al.*, 2010; Speier, 2006). As a result, in this study, we operationalize the performance as users' perceived decision quality and perceived decision effort, the two central components of the effort–accuracy framework.

2.3.2 Effort–accuracy framework. The effort–accuracy framework (Payne *et al.*, 1990) is often used in understanding the outcome of the decision process in decision support systems. Silver (1991) stated that decisional guidance and system restrictiveness are the attributes of designing decision support systems, which affect the cognitive process. Effort–accuracy framework relates to limitations in human cognitive capacity. It was developed in the behavioral decision-making field, where the previous researches indicated that effort is an important factor in strategy selection in the decision-making process (Häubl and Murray, 2006; Qiu and Benbasat, 2009; Xiao and Benbasat, 2007). According to this framework, when an individual needs to make a decision, he/she is concerned not only about the decision quality but also the perceived decision effort (Qiu and Benbasat, 2009; Xiao and Benbasat, 2007, 2018). Among the decision strategies at hand, a person will choose a strategy that can create the best trade-off between the expected accuracy obtainable from strategy application and the cognitive effort required to complete it (Xiao and Benbasat, 2007; Xu *et al.*, 2014b). In other words, the primary objectives of an individual are to maximize accuracy and minimize cognitive effort (Wang and Benbasat, 2012; Xiao and Benbasat, 2007). Previous studies often regard accuracy as an element of decision quality or decision outcome (Chu and Spires, 2003; Todd and Benbasat, 1991). Wang and Benbasat (2009) illustrated that effort is more important than accuracy or perceived advice quality affecting the ACA usage intention. In the online context, ACAs help companies collect information and personalize choices based on consumer's preferences. Developers are improving the innovation function of ACAs, which leads to the expansion of information processing boundary, then increases the accuracy and reduces effort, afterward affects the ACAs engagement or using continuance, and positively influences the purchase intention.

In summary, the effort–accuracy framework proposes that people would go through three stages of decision evaluations: based on the decision aids available at hand (1) users would use their cognitive and behavioral assessments (2) to evaluate the decision aids in terms of decision quality and efforts (3) (Wang and Benbasat, 2009). Similarly, the cognitive fit theory argues that people would base on the information presentation and the task at hand (1) to perceive the mental representation as cognitive fit (2) and evaluate their problem-solving performance (Shaft and Vessey, 2006). Either the cognitive fit theory or the effort–accuracy framework could be used to explain part of what the present paper proposes, however, only the combination and adaptation from both theory and framework could produce the comprehensive explanation that we search for. Thus, this paper utilizes both the cognitive fit theory and the effort–accuracy framework as the theoretical framework.

2.3.3 Problem-solving task as shopping task. Tasks are defined as a person's actions that start with the inputs, go through a series of actions and lead to the outputs (Goodhue and Thompson, 1995). Problem-solving or decision process is one of those tasks (Hong *et al.*, 2004, 2007). In information retrieval research, users could acquire the information online either via searching or browsing tasks (McDonald and Chen, 2006). E-commerce literature has also distinguished these searching and browsing tasks as two common types of online shopping tasks, corresponding to online shopping behaviors of consumers (Adipat *et al.*, 2011; Hong *et al.*, 2004; Yin and Zhai, 2006). In searching tasks, customers search for a known, particular product or brand for their online shopping purpose, it is usually considered as "known-item searching," while in browsing tasks, users are indecisive about what they want to buy and normally browse any product or brand (Hong *et al.*, 2007). Hence, browsing tasks are less specific than searching tasks. For those who need to complete browsing tasks for a supposed product, they have to fix their personal preferences for product assorted characteristics, create a seeking plan to evaluate the alternatives, then employ some decision strategies or methods to make the purchase decision (Hong *et al.*, 2004, 2007). If consumers have distinct tasks in their minds, they will act in different ways, consequently, the sellers should offer them different decision support aids or tools (Yan and Chen, 2015).

2.4 Hypothesis development

2.4.1 Decisional guidance. Guidance is provided in the form of suggestions and information to support users in their decision-making, problem-solving or task completion process using the systems (Hardin *et al.*, 2017; Morana *et al.*, 2014a). Suggestive guidance gives users distinct recommendations on how to perform their discretion (Morana *et al.*, 2014a; Silver, 2006). This guidance provides very explicit descriptions and suggestions on how to progress the decision so that it can enhance the decision quality, raise user satisfaction and decrease the time and effort required for the decision-making process (Al-Natour *et al.*, 2011). Informative guidance provides appropriate information for users to choose without recommending or suggesting how to act (Morana *et al.*, 2014a; Silver, 2006). This type of guidance provides very clear knowledge and empowers user's understanding (Arnold *et al.*, 2006; Morana *et al.*, 2014b).

The cognitive fit theory states that when figuring out a decision-making problem (for example, buying a product online), the decision-making performance of an individual will be affected by the fit between how the information is demonstrated to solve that problem (for example, product information representation) and the essence of such problem (for example, shopping behavior or shopping task required of customers) (Kelton *et al.*, 2010; Speier, 2006). Shaft and Vessey (2006) in their research found that when the individual working in the circumstance that produced a cognitive fit, their performance of completing the modification task was improved. Review research about cognition of Davern *et al.* (2012) illustrated that problem-solving tasks ranged from simple to complex, however, the performance was good or bad depending on not only how complex the task was but also the design difference. That recognition of different designs produced different results for different tasks come up with

the notion of cognitive fit between information presentation and task representation (Chan *et al.*, 2012; John and Kundisch, 2015; Kelton *et al.*, 2010).

Previous studies have paid attention to find the best fit between information presentation and tasks (Speier, 2006). Hong *et al.* (2004) found that for a browsing shopping task, the list format showed a better fit, and a matrix format of information facilitated the searching shopping task better to improve the task performance. Browsing tasks include more complicated cognitive actions, such as attribute evaluation and alternative comparison, thus require specific guidance as suggestive guidance. Meanwhile, searching tasks only ask for a quick scan to find a specific product, therefore, clear information of those target products, without any recommendation, would be more helpful (Hong *et al.*, 2007). Based on the above literature, we develop the first hypotheses as follows.

H1a. When performing the searching task, the informative guidance leads to a greater perceived cognitive fit than suggestive guidance.

H1b. When performing the browsing task, the suggestive guidance leads to a greater perceived cognitive fit than informative guidance.

2.4.2 Communication style. Previous studies in human behavior focused on the importance of interpersonal communication and investigate the communication elements and how they influence the interaction outcomes (Keeling *et al.*, 2010; Kim *et al.*, 2011; Verhagen *et al.*, 2014; Yang *et al.*, 2017). Communication style refers to "the format, ritual or mannerism which the buyer or seller used in the interaction" (Yang *et al.*, 2017) and different agents with different communication styles will create different adaptation among customers (Köhler *et al.*, 2011). Previous research found that when customers do not have enough knowledge and problem solutions, they will judge the service based on the communication style of the providers or sellers (Verhagen *et al.*, 2014).

Communication style could be categorized as task-oriented or social-oriented (Dabholkar *et al.*, 2009; Verhagen *et al.*, 2014). The social-oriented communication style implicates the cues of empathy, friendliness and personality in a conversation. On the other hand, a task-oriented communication style concentrates on a well-structured and goal-oriented conversation. Previous marketing researches indicated that there were differences in recipients' mental cognition, affectivity and behavior between those two kinds of communication style (Dabholkar *et al.*, 2009; Keeling *et al.*, 2010; Van Dolen *et al.*, 2007). Verhagen *et al.* (2014) found that a CA with social-oriented communication style created the sensitive and sociable feelings of human contact, more social presence rather than the task-oriented communication style, and a closer customer relationship because they were more "social-emotional in nature." Cognitive fit indicates design customization to different specific tasks. Jiang and Benbasat (2007) found that people have the intention to find the information presentation formats which can enhance their performance in decision-making and provide a greater fit between such presented information and the type of tasks. Communication style is another important factor to create that better fit.

Information seeking, either searching by browsing or searching by queries (Olston and Chi, 2003), in general, is considered a challenging task that requires a support system (Marchionini and Shneiderman, 1993). To the best of our knowledge, there is no clear conclusion on which type of information searching, either search for a specific product or browse for information, is more challenging. Compared to the well-defined searching task, browsing is "appropriate for ill-defined problems and for exploring new task domains" (Marchionini and Shneiderman, 1993, p. 71). To find the needed information, users would utilize different information-seeking strategies, for example, users who search for specific information would use goal-directed strategies while for browsing, the wandering strategy would be more appropriate but with longer seeking time (Marchionini and Shneiderman, 1993). Janiszewski (1998) found that a goal-directed search is faster because the target is clear

and less distracted by nonfocal information. On the contrary, exploratory search (browse) is slower because users could be distracted by nonfocal information. Browsing task would also require more specific information after each seeking round (White and Roth, 2009).

Within this research, we would like to investigate the searching and browsing tasks that lead to a final decision with the help of chatbots in the e-commerce context. Therefore, we argue that searching for a specific product would be done quickly while browsing would take a longer time as well as more information to complete the task. Customers who have a browsing task would need to do more analytics to come to a final decision, thus, would focus more on solving the task at hand and prefer a less distractive communication style like task-oriented. While for searching task, consumers know what to search for, thus will be more open to small talks and friendly talk with an agent would generate a better cognitive fit. Based on the above literature and arguments, we develop the next hypotheses.

H2a. When performing the searching task, the social-oriented communication style leads to a greater perceived cognitive fit than task-oriented communication style.

H2b. When performing the browsing task, the task-oriented communication style leads to a greater perceived cognitive fit than social-oriented communication style.

Associated with the above-mentioned relationship between decisional guidance and shopping tasks, and the relationship between communication style and shopping tasks presented above, we argue that for searching task where consumers know what they are looking for as well as more relaxed to enjoy the interactions with sellers, informative guidance and social-oriented communication style would generate the better cognitive fit. In contrast, for browsing task where consumers are uncertain of what they need and more focused on the task, suggestive guidance and task-oriented would facilitate better. Thus, we propose the following hypotheses.

H3a. When performing the searching task, the interaction of informative guidance and social-oriented communication style will lead to a greater perceived cognitive fit.

H3b. When performing the browsing task, the interaction of suggestive guidance and task-oriented communication style will lead to a greater perceived cognitive fit.

2.4.3 Perceived decision quality. The ACA application to support consumers to complete their shopping task affects their decision-making processes and outcomes. Consumers will achieve greater decision quality and can reduce their decision effort with agents' support (Qiu and Benbasat, 2009; Wang and Benbasat, 2012; Xiao and Benbasat, 2007). Previous studies measure decision quality using outcomes, such as consensus (Cooper and Haines, 2008), quality (McNamara *et al.*, 2008), satisfaction (Drury-Grogan *et al.*, 2017), confidence (Sanchez *et al.*, 2015; Schwarz and Schwarz, 2007), accuracy (Speier and Morris, 2003; Zhu *et al.*, 2014). The cognitive fit theory also stated that in graph/table presentation researches, they usually explored the decision outcomes like performance, interpretation accuracy and confidence in the acquired results or decision-making satisfaction (Vessey, 1991). Perceived accuracy refers to "the degree to which a consumer believes that an agent would recommend accurate product information" and is a very important indicator to measure decision quality (Zhu *et al.*, 2014). Decision confidence refers to customers' thoughts about the decision they have made. The higher the decision confidence they get, the higher belief and trust strength in their decision (Sanchez *et al.*, 2015; Schwarz and Schwarz, 2007).

As both perceived accuracy and decision confidence are essential elements, we decide to choose these two determinants to measure the perceived decision quality in this current research. We expect that the higher cognitive fit between the design feature of the ACA and the shopping task can lead to the higher decision quality perception of customers, thus we propose the following hypothesis:

H4. Perceived cognitive fit between ACA information format and shopping task positively influences perceived decision quality of customers.

2.4.4 Perceived decision effort. In the context of online shopping, consumers' decision efforts are related to how much effort consumers need to pay in the process of acquiring information, assessing alternatives and coming up with a product choice (Xiao and Benbasat, 2007). Based on the preferences customers had revealed, recommendation agents (RAs) could absorb the sorting process, help customers reduce their information and alternative searching, consequently, reduce their effort. Users enjoy the interaction created during the decision-making processes that mobile RAs provide, so they intend to implicate more in the decision-making process (Lee and Benbasat, 2010). The active engagement of mobile RA users in decision-making work makes them depreciate the time and effort they spend (Komiak and Benbasat, 2006; Lee and Benbasat, 2010). RAs could also confine the alternatives for customers, increase the fit between the agents' presentation and the decision tasks, as a result, reduce the efforts (Häubl and Murray, 2006). In this study, we expect that the higher cognitive fit between the design feature of the ACA and the shopping task can lead to the lower decision effort perception of the customer, thus we propose the following hypothesis:

H5. Perceived cognitive fit between ACA information format and shopping task negatively influences perceived decision effort of customers.

2.4.5 Perceived task complexity and its moderating effect. Different tasks have different complexity levels and each level is a reflection of an individual's assessment of cognitive load (Campbell, 1988). Similarly, cognitive load theory posits that an individual's cognitive load is affected by the cognitive resources they dedicate to the task and the associated attentional demands with the task (Sweller, 1988). It means that when the task complexity increases, decision-makers will experience higher cognitive loads and then they need to find a way to increase the decision accuracy to create the trade-off with the decision effort.

Task complexity is an important determinant to affect the effectiveness of various service technologies (Gupta *et al.*, 2013; Jiang and Benbasat, 2007; Speier and Morris, 2003; Xu *et al.*, 2014a). Furthermore, the task of purchasing online products has a wide range of complexity, depending on the complexity of product attributes. Xiao and Benbasat (2007) proposed that "product complexity moderates the effects of RAs use on users' decision quality and decision effort." They stated that the RA used for highly complex products leads to a greater raise in decision quality and greater reduction of decision effort than for low complex products. Therefore, the advantages of using RAs to increase the decision quality and decrease the decision efforts will be greater for a more complex product (Xiao and Benbasat, 2007). When task complexity increases, the agents will need to provide more product attributes and information, then product understanding of customers will be improved, as a result, increase the agent's performance (Jiang and Benbasat, 2007). Adipat *et al.* (2011) also found that the more the complexity became, the more the impact of information presentation on users' performance increased. It means that when the task is more complex, the cognitive fit between the information format displayed by the ACA and the shopping task will show more effectiveness in enhancing decision quality and reducing decision effort. Or in other words, when the task complexity is low, consumers can easily solve the task with the help from agents. When the complexity increases, consumers might need more support to solve the task at hand. In this case, the fit between the information presentation and the task plays a more important role in helping consumers to make a precise decision as well as reduce the decision-making time and effort. Thus, we propose the two moderating hypotheses as below.

H6a. Perceived task complexity strengthens the relationship between perceived cognitive fit and perceived decision quality.

H6b. Perceived task complexity strengthens the relationship between perceived cognitive fit and perceived decision effort.

Figure 1 below illustrates the proposed research model for this study.

3. Methodology

3.1 Experimental design

We conducted a 2x2x2 full factorial between-subject online field experiment on a manipulated Facebook fanpage to study the effects of decisional guidance, communication style and shopping task on consumers' decisional process. According to Facebook in early 2019, there are more than 2.32bn monthly active users all over the world and over 300,000 active Facebook Messenger Chatbots. Furthermore, the results of a survey from Oracle (2016) showed that among 800 business leaders and decision-makers interviewed, 80% already used or planned to use chatbots by 2020. This data shows the importance of chatbots on the Facebook platform; thus, we chose Facebook fanpage and Facebook Messenger Chatbots as our experiment context. Our experiment design is shown in Table 1.

3.2 Setting selection and factor manipulation

For setting selection, a presurvey was conducted to select the product category for the experiment. The initial list of products was selected based on previous e-commerce studies (Xiao and Benbasat, 2014). The final list included book, digital versatile disc (DVD), camera, smartphone, fashion clothing, sports clothing, hotel booking, food and beverage and grocery. Participants were asked to rate that list of products and services as their interest level,

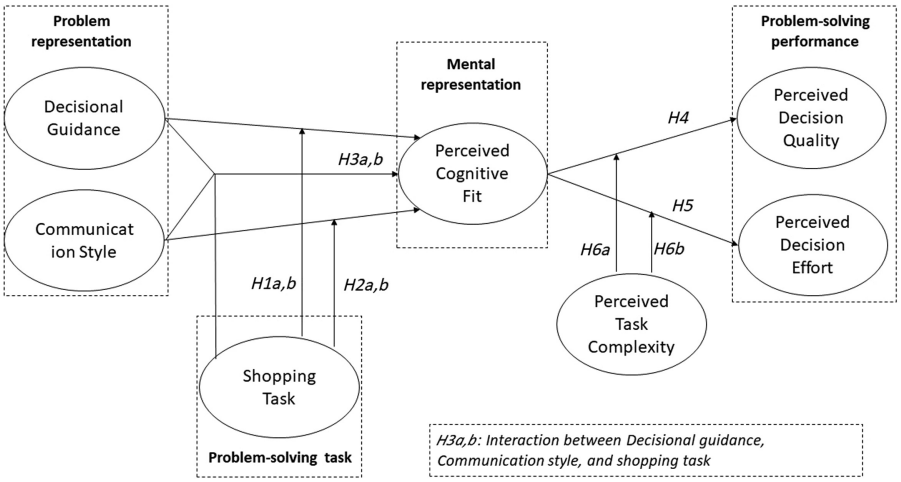

Figure 1. Research model

Decisional guidance		Suggestive guidance		Informative guidance	
Communication style		Task-oriented	Social-oriented	Task-oriented	Social-oriented
Shopping task	Searching	SST (n = 31)	SSS (n = 33)	SIT (n = 32)	SIS (n = 30)
	Browsing	BST (n = 31)	BSS (n = 30)	BIT (n = 32)	BIS (n = 30)

Table 1. Experimental design

shopping frequency, and the need for an online shopping assistant. To avoid the bias and compounding effects, the chosen product for the experiment must satisfy certain criteria, including no different interest between gender and ages, with the average interest level and shopping frequency, and the high need for a virtual assistant to support customers' purchasing decision. A sample of 65 random Facebook users took part in the presurvey (35 females, aged between 18 and 34). Camera, sports clothing, and food and beverage satisfied the set requirements. To prevent the effect of product interest and purchasing frequency, the products which have the neutral means should be chosen. The chosen product should also have a high agent-necessary level, so the chatbot could amplify its utilization. Thus, sport clothing products were selected to design our experiment. Each factor for manipulation was then conceptualized and operationalized for the experiment.

First, for *suggestive guidance condition* of the decisional guidance, the chatbot was designed to offer some recommendations to customers about which style and color to choose. For *informative guidance conditions*, the chatbot was designed to offer customers very clear product information without recommending them to any specific product to choose from. Second, for communication style, according to Dabholkar *et al.* (2009), because "*task-oriented communication style* is goal-oriented, purposeful, and structured," the chatbot was designed to only provide the necessary information and focus on the efficiency for this condition. Malinowski (1994) stated that small talk is one of the major determinants of social-oriented communication style because it creates a close relationship between communication parties. Small talk could be a compliment (e.g. *You are so nice! That's great!*) or an expression of agreement (e.g. *You are right*). Social-oriented communication style could also show empathy by using "emoji," which is some symbols used to express emotions, e.g. 👏 🎉). Finally, for shopping task, those assigned to the searching condition were instructed to go to the chatbot with "the goal of efficiently finding something specific within the bot," which means they were given the task of purchasing a specific product. Those assigned to the browsing condition were instructed to "looking at whatever you consider interesting and/or entertaining." Detailed product information, including price, product image, material and other attributes, was available on the chat window with the chatbot.

Associated with the attributes of sports clothing, we designed the fanpages and the chatbot's conversations and factorial manipulation according to the conceptualization and operationalization as shown in Table 2 below.

We created eight different Facebook fanpages, equivalent to eight experimental treatments. All fanpages had the same information about the sports clothing including general information of the shop, photos and albums of sports clothing offered by the shop, as well as the appearance of the fanpages. According to the treatment it belonged to, each fanpage was accompanied by one chatbot with specifically designed features. In searching task conditions, we had four chatbots to four respective conditions of conversation design as suggestive guidance and task-oriented communication style, suggestive guidance and social-oriented communication style, informative guidance and task-oriented communication style, and informative guidance and social-oriented communication style. Those four chatbot designs were replicated for browsing task conditions. We had eight chatbots corresponding to eight treatments in total. Examples of chatbot conversations are provided in Appendix 1.

To check the success of the manipulation of independent variables, we generated six questions and asked the respondents after they were assigned and exposed to the message and situation. There were two scenarios for decisional guidance stimuli, categorized by suggestive guidance and informative guidance. We checked the manipulation by asking participants to indicate their level of agreement toward two questions: "The chatbot recommended me the best seller style that I should choose" (seven-point Likert scale, from 1–strongly disagree to 7–strongly agree) and "The chatbot only listed all styles they had rather than giving a clear recommendation of what I should choose" (True or False). Communication

Variables	Manipulation	Conceptualization	Sources	Operationalization
Decisional guidance	Suggestive guidance	"Makes explicit recommendations to the user on how to exercise his or her discretion"	Silver (2006)	Suggestions of styles and colors were given
	Informative guidance	"Provides pertinent information that enlightens the user's choice without suggesting or implying how to act"		Necessary product information was provided
Communication style	Task-oriented	"Focuses on goals, exchanging knowledge, and fulfilling recipient needs in terms of information"	Chattaraman *et al.* (2019); Dabholkar *et al.* (2009)	Communication style focused on the efficiency
	Social-oriented	"Focuses on relationships, satisfying the emotional needs of recipients, and facilitating interactions"		Small talks and emoji were added to the communication
Shopping task	Searching Task	Consumers have particular and specific objectives	Hong *et al.* (2004)	The task was to search for a specific product
	Browsing Task	Consumers only have general objectives		The task was to search for a product that the participant likes

Table 2. Factor manipulation

style was manipulated by task-oriented style and social-oriented style. We checked the manipulation by asking participants to indicate their level of agreement toward two questions: "The chatbot used small talk (e.g. Great! Come on!) with emoji (e.g. 👏 🤙) in the conversation" (seven-point Likert scale, from 1–strongly disagree to 7–strongly agree) and "The chatbot used plain text without emoji in the conversation" (True or False). The shopping task was stimulated as searching task and browsing task. We checked the manipulation by asking participants to indicate their level of agreement toward two questions: "Before chatting with the chatbot, my purpose was:" (Browsing for a general product or Searching for a very specific product) and "Before chatting with the chatbot, my task was looking for a very specific product" (seven-point Likert scale, from 1–strongly disagree to 7–strongly agree).

3.3 Participants and measurements

We chose to focus on people with ages ranging from 18–34 years old as this age group spends a longer time online and has good computer self-efficacy. The participants were invited via email, social networks and direct messaging. To encourage participation, two couple of movie tickets were given to two random participants as a lucky prize. The measurement items for dependent constructs were adapted from previous studies and measured using a seven-point Likert scale ranging from 1 (strongly disagree) to 7 (strongly agree). Definitions of all variables as well as how they were conceptualized in this study are shown in Table 3 below, detailed measurements as Appendix 2.

3.4 Tasks and experimental procedures

Participants were given a task to look for a sports clothing product as a gift for a friend. Following previous studies (Lee and Benbasat, 2011; Wang and Benbasat, 2007, 2016; Xu *et al.*, 2014b), the gift-giving task was chosen to avoid their personal preference and

Variables	Adapted definition	Sources	No of items	Sources
Decisional guidance	"How an automated conversational agent enlightens or sways its users as they structure and execute their decision-making processes"	Hardin *et al.* (2017)	Manipulated factors	
Communication style	"The format, ritual or mannerism in communication which conversational agent used to interact with customers"	Yang *et al.* (2017)		
Shopping task	"A piece of work that a customer performs through a sequence of actions to produce the outcomes that compatible to the shopping goal"	Goodhue and Thompson (1995)		
Perceived cognitive fit	"The match between the information emphasized in the representation type and that required by the task type"	Vessey (1991)	6	Garaus *et al.* (2015); Lu and Yang (2014)
Perceived task complexity	"An individual's perception of how complex the task is"	Gupta *et al.* (2013)	4	Gupta *et al.* (2013)
Perceived decision quality	"The extent to which a consumer feels confident in the quality of the product choice she has made at an online platform"	Xiao and Benbasat (2018)	7	Lee and Benbasat (2011); Xiao and Benbasat (2018)
Perceived decision effort	"A consumer's assessment of the cost associated with evaluating and choosing products at an online platform"	Xiao and Benbasat (2018)	4	Xiao and Benbasat (2018)

Table 3.
Variables definitions

potential biases in their evaluations as they buy for themselves. To complete the task, participants were asked to go to a Facebook fanpage that sold sports clothing online and had a conversation with the chatbot.

The experimental procedures consisted of five steps. First, the participants must state their consent. Second, we introduced ACA and how Facebook Messenger Chatbot worked to the participants. Third, we assigned the online shopping task and gave the instructions to complete the task. Fourth, the participants were randomly assigned to one of eight fanpages, they needed to browse the products offered by the shop, then messaged the fanpage to ask for support by clicking the "Send Message" button. All eight chatbots (one for each experimental condition) were performed online and participants were able to access and chat with one of them using Facebook Messenger. The task finished when the participants clicked "Purchase" at the end of the chat. To make sure the participants have enough time to interact with the chatbot, we designed the chatbots so that it took at least five minutes for the participants to complete one round of conversation. If the participants had not satisfied with the suggested products, instead of clicking the "Purchase" button, they could go back to the previous menu and continue searching or browsing until they decided to purchase. In this case, the interaction time would be longer depending on their chatting time. Finally, the participants were asked to complete a questionnaire that included the manipulation check questions, the variable measurements and background questions to collect their demographic information.

Before the main experiment, we conducted a pilot test with 39 participants. They were international graduate students who were active on Facebook, had online shopping

experience, at the age of 18–34. Via the pilot test, we have checked the feasibility and flow of the experimental procedures, the validity of our manipulation and manipulation checks. Based on the pilot test results, we adjusted and improved our experimental design.

3.5 Control variables

Since our focus is on the effects of the quality of chatbot design on consumer online purchase decision-making performance, we included a set of control variables which might affect the consumer decision process to rule out rival explanations of online product decision-making performance. Prior research illustrated the effect of customers' online shopping experience differences in their decision process (Melis *et al.*, 2015). The more experience they have in online shopping, the more familiar they feel and this leads to a more efficient decision (Yoon *et al.*, 2013). Besides, during the online shopping process, acquiring product information is very important, so that customers who have different self-efficacy levels in acquiring information might have a different perception of decision quality (Pavlou and Fygenson, 2006). High self-efficacy could reduce the information load (Zha *et al.*, 2013) and increase performance expectations or online purchasing behavior (Luszczynska *et al.*, 2005). Finally, Lee and Benbasat (2011) indicated that when users perceive that they have high control over RAs' execution, they will believe that they can make the agent give recommendations that reflect their preferences so they will judge recommendations of such agents more accurate. Therefore, beside the demographic information (age, gender, Facebook usage frequency), online shopping experience, self-efficacy and perceived control were included as control variables.

4. Results

4.1 Demographics and sample size

The main experiment was conducted online with a total useable sample of 249, among which there were 101 males (40.6%) and 148 females (59.4%). Most of the participants were aged between 18 and 34 (94.8%). Table 4 shows the demographic characteristics of all participants. The result of the random assignment is included in Table 1, showing each of the eight treatments has 30 to 33 samples. Thus, the size is considered to be equal among treatments.

4.2 Manipulation check and factor analysis

Before analyzing the collected data, we performed the manipulation check to verify the success of our manipulation. We conducted 6 one-way analysis of variance (ANOVA) analyses using Statistical Product and Service Solutions (SPSS) 18 based on 6 manipulation check questions, 2 questions for each manipulated factor. The ANOVA results showed that there were significant differences between compared groups (suggestive vs informative guidance, task-oriented vs social-oriented communication style and searching vs browsing task). Thus, we concluded that our manipulation was successful.

Factor analysis and reliability tests were then conducted using SPSS 18 to check the validity of our measurement model, including perceived cognitive fit, perceived task complexity, perceived decision quality and perceived decision effort. Among the items, FIT2 was deleted because of low factor loading. As illustrated in Table 5, all remaining items had loadings greater than 0.7 and Cronbach's alpha greater than 0.8, indicating the remaining measurement model had good validity and reliability (Hair *et al.*, 2010). We thus moved on with hypothesis testing.

4.3 Analysis of variance

We conducted ANOVA tests to examine the interaction effects between decisional guidance, communication style and shopping task on perceived cognitive fit. Results for ANOVA were

Table 4. Characteristics of main experiment respondents

Demographics	Category	Frequency (n = 249)	Percentage (%)
Gender	Male	101	40.6
	Female	148	59.4
Age	18–24	53	21.3
	25–34	183	73.5
	Above 34	13	5.2
Amount of time spending on Facebook per day	<1 hours	49	19.7
	1–3 hours	99	39.8
	3–5 hours	63	25.3
	5–7 hours	26	10.4
	>7 hours	12	4.8
Online shopping frequency on average (times per month)	<1	80	32.1
	1–3	125	50.2
	3–5	24	9.6
	>5	20	8.0

Table 5. The results of factor analysis and reliability test

Construct	Items	Mean	S.D.	Factor loading	Eigen value	Cumulative explained variance	Item-to-item correlation	Cronbach's alpha (α)
Perceived cognitive fit	FIT1	5.31	1.333	0.834	3.459	69.17%	0.731	0.888
	FIT3	5.22	1.225	0.820			0.712	
	FIT4	5.33	1.249	0.843			0.744	
	FIT5	5.53	1.205	0.806			0.694	
	FIT6	5.29	1.327	0.855			0.761	
	FIT2	5.34	1.260	Deleted	–	–	–	–
Perceived task complexity	TACO1	3.75	1.832	0.900	2.982	74.54%	0.723	0.885
	TACO2	4.09	1.803	0.898			0.674	
	TACO3	3.76	1.900	0.844			0.807	
	TACO4	3.44	1.913	0.808			0.803	
Perceived decision quality	QUA1	5.16	1.439	0.797	4.507	64.39%	0.714	0.906
	QUA2	5.05	1.425	0.802			0.720	
	QUA3	4.91	1.418	0.752			0.664	
	QUA4	5.27	1.266	0.859			0.793	
	QUA5	5.02	1.428	0.823			0.746	
	QUA6	4.89	1.497	0.845			0.774	
	QUA7	5.08	1.458	0.730			0.639	
Perceived decision effort	EFF1	3.21	1.759	0.789	2.909	72.73%	0.645	0.874
	EFF2	3.10	1.784	0.872			0.759	
	EFF3	2.97	1.752	0.879			0.768	
	EFF4	2.70	1.692	0.867			0.751	

summarized in Table 6. Hypothesis H1a stated that for searching tasks, informative guidance would lead to a better fit than suggestive guidance. However, the results showed the opposite and the difference in the fit among the two groups was not significant ($F = 1.606, p > 0.1$), thus H1a was not supported. H1b argued that for browsing tasks, suggestive guidance would generate a better fit. With $F = 2.950$ and $p < 0.1$, suggestive guidance generated a better fit than informative guidance (mean of perceived cognitive fit generated by suggestive guidance was 5.367 compared to 5.054 of informative guidance), we concluded that H1b was supported at $p < 0.1$ level. H2a stated that for searching task, social-oriented communication style was more appropriate than task-oriented. Our results showed that social-oriented communication

Independent variable	Treatment	*N*	Mean	S.D.	*F*	*p*
Main effect						
Shopping task	Browsing	123	5.209	1.016	5.211	0.023
	Searching	126	5.504	1.084		
Decisional guidance	Informative	124	5.217	1.113	4.416	0.037
	Suggestive	125	5.499	0.987		
Communication style	Task-oriented	126	5.158	1.041	9.442	0.002
	Social-oriented	123	5.564	1.043		
Shopping task*Decisional guidance	Browsing*Informative	62	5.054	1.017	2.950	0.088
	Browsing*Suggestive	61	5.367	0.998		
	Searching*Informative	62	5.380	1.188	1.606	0.207
	Searching*Suggestive	64	5.625	0.968		
Shopping task* Communication style	Browsing*Task-oriented	63	5.146	0.938	0.505	0.478
	Browsing*Social-oriented	60	5.276	1.096		
	Searching*Task-oriented	63	5.171	1.141	13.043	0.000
	Searching*Social-oriented	63	5.838	0.918		
Shopping task *Decisional guidance* Communication style	Browsing*Informative* Task-oriented	32	5.150	0.884	3.996	0.047
	Browsing*Informative* Social-oriented	30	4.953	1.150		
	Browsing*Suggestive* Task-oriented	31	5.141	1.006		
	Browsing*Suggestive* Social-oriented	30	5.600	0.951		
	Searching*Informative* Task-oriented	32	4.968	1.291		
	Searching*Informative* Social-oriented	30	5.820	0.894		
	Searching*Suggestive* Task-oriented	31	5.380	0.938		
	Searching*Suggestive* Social-oriented	33	5.854	0.952		

Table 6. The results of ANOVA (Dependent variable: perceived cognitive fit)

style led to a much greater fit than task-oriented (mean of 5.838 vs 5.171), thus H2a was supported. However, for browsing task, both communication styles generated almost the same fit. Therefore, H2b was not supported.

Finally, for the interaction between the information presentation of decisional guidance and communication style and shopping tasks, H3a argued that for searching tasks, the interaction between informative guidance and social-oriented communication style would lead to greater fit. The results showed that social-oriented communication style while interacting with either suggestive or informative guidance would create the same better fit (mean of 5.854 and 5.820 respectively) compared to task-oriented style. Thus, H3a was not supported. H3b argued that for browsing tasks, suggestive guidance and task-oriented communication style would facilitate better. However, according to our findings, it was the combination of suggestive guidance and social-oriented rather than task-oriented style that facilitated better. So, H3b was also not supported.

Even though H3a and H3b were not supported, we found the significant interaction effects between decisional guidance and communication style and shopping tasks. To have a better understanding of the interacting effects, following Yorkston and Menon (2004), we investigated for further understanding of the three-way interaction between decisional

guidance, communication style and shopping task. Based on the results, we fixed the decisional guidance as to either informative or suggestive and explored the interaction between communication style and shopping tasks. When decisional guidance was fixed as informative, the ANOVA results showed a significant two-way interaction ($F = 7.418$, $p < 0.01$). When the chatbots used the social-oriented communication style, the fit was dramatically higher in searching tasks compared to browsing tasks. On contrary, for chatbots using task-oriented communication style, the fit tended to slightly drop from browsing tasks to searching tasks. When the decisional guidance was fixed as suggestive, no significant two-way interaction was detected ($F = 0.02$, $p > 0.9$) and in all cases, the chatbots which used social-oriented communication style generated a better fit and the fit was raised from browsing task to searching task. The interaction is further illustrated in Figure 2.

4.4 Regression analysis

For testing Hypotheses 4 to 6, we used linear regression analysis. We first aggregated the mean scores of the variables, and then calculated the centered value of the mean by subtracting the mean scores by the aggregated mean. The interaction terms were calculated by multiplying the central means of the respective variables. This step was necessary to reduce the possible multicollinearity among the interaction terms (Chang and Chuang, 2011). To have a thorough understanding of the effects on perceived decision quality and perceived decision effort, we conducted linear regression analyses on two dependent variables: perceived decision quality and perceived decision effort. We conducted three different regression models to estimate the effect of perceived cognitive fit (Model 1), perceived cognitive fit, perceived task complexity and interaction effect (Model 2), and perceived cognitive fit, perceived task complexity, interaction effect and control variables (Model 3) on each dependent variable. Table 7 summarizes the variable correlation and Table 8 reports the results of the regressions. The results showed that perceived cognitive fit positively influenced perceived decision quality ($\beta = 0.621$, $p < 0.001$) and negatively influenced perceived decision effort ($\beta = -0.253$, $p < 0.001$); thus, Hypotheses 4 and 5 were supported. For moderating effects, perceived task complexity had little impact on the relationship between perceived cognitive fit and perceived decision quality while significantly strengthened the relationship between perceived cognitive fit and perceived decision effort. In other words, Hypothesis 6a was not supported but Hypothesis 6b was. We further illustrated the significant moderating effect in Figure 3. For the control variables, all control variables showed no effect on the dependent variables except that perceived control had a positive effect on perceived decision quality.

4.5 Mediation analysis

From the results of the variable correlation provided in Table 7, perceived cognitive fit and perceived decision quality were found to be highly correlated (Mukaka, 2012), therefore, it is suggested to conduct the mediation test to further explore the relationship between these two variables. We conducted the Sobel test (Baron and Kenny, 1986) to test the mediating effects of perceived cognitive fit (M) between information presentation (decisional guidance and communication style and problem-solving task (shopping task)) as independent variables (IVs) and decisional performance (perceived decision quality and perceived decision effort) as dependent variables (DVs). First, we ran different linear regressions of IVs on DVs (step 1), IVs on M (step 2) and IVs and M on DVs (step 3). Then we calculated the Sobel test statistics. The results in Table 9 show that perceived cognitive fit partially mediates the relationships between IVs and decision quality, while it has no mediating effect between decisional guidance and perceived decision effort, and between shopping tasks and perceived decision effort. However, perceived cognitive fit fully mediates the relationship between communication style and perceived decision effort.

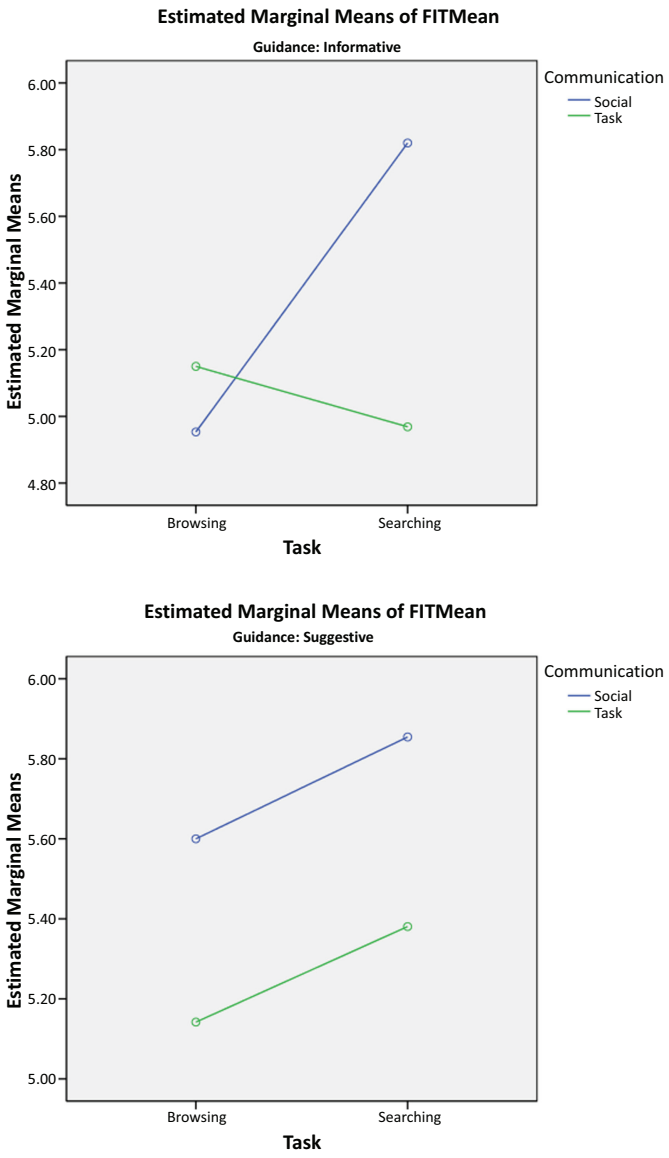

Figure 2.
Plots of three-way interaction

5. Discussions and implications

5.1 Discussions

The results revealed that there are significant differences between the two types of each independent variable (decisional guidance, communication style and shopping task) when affecting the cognitive fit. Compatible to previous studies, our findings show that suggestive guidance has a stronger effect than informative guidance (Parkes, 2013), social-oriented communication style has a higher impact than task-oriented style (Verhagen *et al.*, 2014), and searching task showed a stronger influence than browsing task

Table 7.
Correlation matrix

Constructs	Mean	S.D.	FIT	QUA	EFF	TACO	FITA
FIT	5.359	1.059	1				
QUA	5.054	1.136	0.734***	1			
EFF	2.996	1.488	−0.127*	−0.088	1		
TACO	3.760	1.607	0.171**	0.144*	0.554***	1	
FITA	0.290	1.712	−0.135*	−0.051	0.327***	0.287***	1

Note(s): (1) Mean = average scores except for FITA (mean-centered scores); S.D. = standard deviation, (2) FIT = Perceived cognitive fit; QUA = Perceived decision quality; EFF = Perceived decision effort; TACO = Perceived task complexity; FITA = FIT x TACO,
(3) All coefficients are calculated based on mean-centered scores,
(4) $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$

Table 8.
The result of main effects and moderating effects

	Perceived decision quality			Perceived decision effort		
Dependent variables	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3
Perceived cognitive fit	0.734***	0.739***	0.621***	−0.127*	−0.201***	−0.253***
Perceived task complexity		0.004	−0.031		0.547***	0.532***
Fit x Taco		0.047	−0.004		0.143**	0.132*
Control variables						
Age			0.044			−0.022
Gender			−0.070			−0.029
Facebook Usage Frequency			0.000			0.083
Online shopping Experience			−0.033			0.092
Self-efficacy			0.031			−0.019
Perceived Control			0.265***			0.082
R^2	0.538	0.541	0.604	0.016	0.375	0.401
Adjusted R^2	0.537	0.435	0.589	0.012	0.367	0.378
VIF Range	1.000	1.071–1.146	1.053–1.431	1.000	1.071–1.146	1.053–1.431

Note(s): $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$

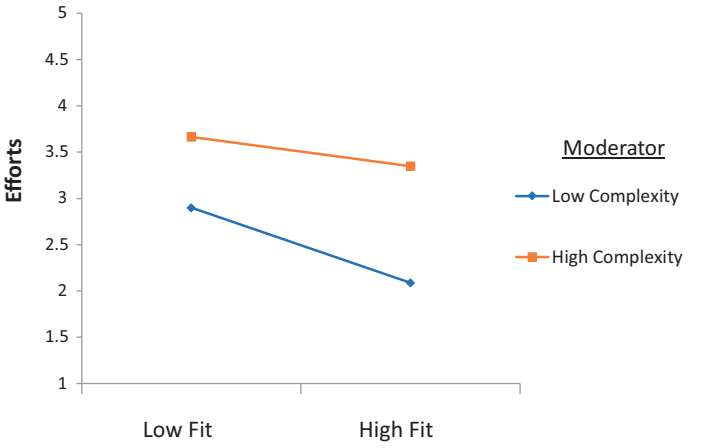

Figure 3.
Plots of moderation effects of perceived task complexity

Table 9. The result of mediation analysis using Sobel test

Independent variable (IV)	Mediator (*M*)	Dependent variable (DV)	Coefficients in regressions IV-DV	IV-M	M-DV	Sobel test statistic	Mediating
Decisional guidance	Perceived cognitive fit	Perceived decision quality	0.416*** (0.054)	0.385*** (0.051)	0.787*** (0.046)	6.906***	Partially
Decisional guidance		Perceived decision efforts	0.065 (0.079)	0.385*** (0.051)	−0.178* (0.089)	−1.933	None
Communication style		Perceived decision quality	0.554*** (0.047)	0.505*** (0.045)	0.787*** (0.046)	9.383***	Partially
Communication style		Perceived decision efforts	−0.072 (0.077)	0.505*** (0.045)	−0.178* (0.089)	−1.968*	Full
Shopping tasks		Perceived decision quality	0.251*** (0.060)	0.253*** (0.056)	0.787*** (0.046)	4.368***	Partially
Shopping tasks		Perceived decision efforts	−0.034 (0.082)	0.253*** (0.056)	−0.178* (0.089)	−1.828	None

Note(s): $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$. Coefficients in regressions are unstandardized values and standard errors are in the brackets

(Yin and Zhai, 2006). The match between information format and task significantly led to the cognitive fit, which once again confirmed the cognitive fit theory. This fit in cognition improved the decision-making performance, not only enhanced the quality of decision but also reduced the effort of the customers. For searching task, while clear suggestions rather than all information provision from the chatbot only generated a marginally higher cognitive fit (not supporting H1a), consumers would perceive higher fit if they communicated with a friendly chatbot (supporting H2a). For browsing task, consumers would prefer the chatbot to give more specific suggestions and guide them through the shopping task rather than giving general information (supporting H1b). Since consumers are more focused on solving the task, the language of the chatbot was not a concern, thus, either social-oriented or task-oriented communication style made no difference to the consumers (not supporting H2b). Even though both H3a and H3b were not supported, our findings also suggested that in most cases, social-oriented communication style dominated, and such domination became clearer when interacting with suggestive guidance and shopping tasks. This dominant effect of social-oriented communication style shows the demand of customers to create a relationship with the ACA, as with a human agent. When interacting with such ACAs, besides the goal of making a purchase decision, customers also want to enjoy the chatting process.

Compatible with the cognitive fit theory and the effort–accuracy framework, the findings show that the cognitive fit between the shopping task and the information format of ACA enhances the decisional outcomes: increases the decision quality of customers and reduces the decision effort. Thus, Hypotheses 4 and 5 were supported. However, the variance explained of perceived decision effort is low, approximately 2%. The definition of perceived decision effort adapted from the research of Wang and Benbasat (2009) is "the amount of effort exerted by the consumer in processing product information and making a choice decision." In this product information processing procedure, it could have a possibility of other factors from cognition existed which generate perceived decision effort rather than cognitive fit, leading to the low variance.

The current study tested the moderation effect of task complexity on the relationship between perceived cognitive fit and two dependent variables. Task complexity only moderated the relationship between perceived cognitive fit and perceived decision effort. The result showed that perceived task complexity significantly enhanced the negative effect between perceived cognitive fit and perceived decision effort. It means that if customers perceive the task is complex, they will evaluate the perceived cognitive fit between the shopping task and the ACA's information format more effective in reducing their decision effort. The moderation effect of perceived task complexity on the relationship between perceived cognitive fit and perceived decision quality is not significant, because of the strong effect of perceived cognitive fit on perceived decision quality. This perceived task complexity also showed its role not only as a moderator but also its ability to directly affect the perceived decision effort. When customers perceive a task as complex, they need to exert more cognitive resources, consequently, leading to an increase in the effort required to complete that task.

Perceived control was found to positively affect perceived decision quality only, which is different from Lee and Benbasat (2011)'s research, which indicated that perceived control positively affected perceived decision quality and negatively affected perceived decision effort. It can explain that in ACA condition, especially in social networking sites, customers consider it as a real chatting conversation, so customers are ready to pay more effort to get more control in this conversation. Another reason is the primary of chatbot's conversation is giving customers options, as they just need to click on the button and limit the typing sentence, it helps them to reduce the effort but also decrease their control of the conversation.

5.2 Theoretical contributions and practical implications

5.2.1 Theoretical contributions. This study contributes to the development of the cognitive fit theory. First, previous studies focused on other problem-solving representations, such as graphical versus tabular information presentation format (Speier, 2006), matrix versus list (Hong *et al.*, 2004) or picture and some other forms of information format, this study extends the problem-solving representation of this theory to the decisional guidance and communication style in the context of ACA. Even though the importance of natural language has long been investigated, the conceptualization of decisional guidance and communication style in the ACA context is novel and appropriate. Our conceptualization and findings not only contribute to the cognitive fit literature but also the literature of ACA design, especially the text-based ACA, i.e. chatbots, where the language is the dominant tool of communication.

Second, the integration of the conversation of an ACA and the shopping task of customers influences perceived decision quality and effort through the perceived cognitive fit. It is common in the cognitive fit literature that researchers focused on the direct relationships between the technology features and the decisional performance (Hong *et al.*, 2004; Speier, 2006) rather than investigating the mediating effect of perceived cognitive fit. To the best of our knowledge, previous studies investigated the cognitive fit in a different context rather than the cognitive fit theory (Garaus *et al.*, 2015; Lu and Yang, 2014). Therefore, the findings in the present study set the light on the role of perceived cognitive fit as the mediator and give a more clear explanation about the mechanism through which ACA influences the perceived decision quality and decision effort. Perceived cognitive fit partially mediates the relationships between all three factors of decisional guidance, communication style, and shopping tasks and perceived decision quality, which means that the three factors could directly influence the quality of the decision or indirectly via the mechanism of perceived cognitive fit. These results suggest that the fit between information presentation and the problem-solving task is important in generating good decision-making quality. For perceived decision effort, only communication style has full mediation via the perceived cognitive fit.

The current study also contributes to the decision aid literature. While previous studies focus on the accuracy of chatbot's conversation as its performance in the communication field, users' acceptance of ACA, user's satisfaction, this study examined ACA as one type of decision aids to support the decision-making process of customers, which focuses on the interaction between the customer and the aid, rather than other decision aids like RA or comparison matrix.

5.2.2 Practical implications. This study also provides several implications from a practical perspective. First, the role of ACA as a decision aid was confirmed, which helps the marketers evaluate accurately the importance of ACA when supporting the decision-making process of customers, so they can pay more attention and invest more to develop chatbots. The potential development of ACA on social networks should be considered to create a more effective interaction with customers and promote sales.

Second, marketers can understand the shopping behaviors of customers through the shopping tasks. As a result, they can design their ACA, in terms of both the guidance and communication style, to fit with the users' shopping behavior, to have a more effective decision aid. The dominant effect of suggestive guidance and social-oriented communication style found in this study suggested that marketers should focus on such type of guidance and communication style to help customers get benefits by increasing the decision quality and reducing their decision effort. The dominant role of social-oriented communication style suggests that businesses should create an ACA that possesses friendly conversational skills and is more like a friend than an online assistant.

Moreover, the finding that perceived control positively affects perceived decision quality suggests that ACA developers may need to improve the performance of ACA to increase the perceived control. This result also implies that practitioners should provide different ACA options for users to choose the most appropriate one to satisfy their goals. By providing the

choices for these consumers, they will perceive such ACA is more controllable and then lead to better decision performance.

5.3 Research limitations and suggestions for future research

As other research, our research is not free of limitations. The variance explained in the construct of perceived decision effort is quite low. Even such low variance explained still contributed to the model, it implies that it could have a possibility of other factors existed which generate perceived decision effort besides the perceived cognitive fit, and the results of this study showed that perceived task complexity might be a potential factor. Further research should consider them as the main effect on perceived cognitive effort in the next studies. Moreover, as the definition of perceived cognitive effort focuses on the customer's perception during the purchasing process, the future study also can go further to the psychological side to find out such factors that affect the perceived decision effort.

Although the perceived cognitive fit construct in this study is a contribution to the cognitive fit theory, we borrowed it from the construct of "task-technology fit" in the literature of task-technology fit theory, which was defined as the degree to which technology assists the users in performing their work or coursework, seems too general when adapting to the ACA context. Future research should do a qualitative study to confirm that construct and see whether another measurement is more accurate for the context of ACA.

Another limitation is the experiment time of this research is quite short, only about ten minutes exclude answering the questionnaire, while it went through all consumer-buying processes, beginning with the recognition of a need (shopping task) and continue through information search, evaluation of alternatives, purchase decision and postpurchase evaluation. This short time is not enough to fully experience the ACA's conversation and then evaluate accurately the performance of the decision supported by the ACA. We highly recommend future studies to conduct a longer experiment to have the most accurate results.

Finally, to the best of our knowledge, there is no clear conclusion of which type of information seeking, searching for a specific product, or browsing to find a suitable product, is more difficult to complete. In this research, we argue that browsing without clear queries take more time and effort to complete, especially in the context that task executors must decide at the end. This argument might not hold in other contexts. Therefore, future research could compare the two types of tasks to have better understandings.

References

Adipat, B., Zhang, D. and Zhou, L. (2011), "The effects of tree-view based presentation adaptation on mobile web browsing", *MIS Quarterly*, Vol. 35 No. 1, pp. 99-121.

Al-Natour, S., Benbasat, I. and Cenfetelli, R. (2011), "The adoption of online shopping assistants: perceived similarity as an antecedent to evaluative beliefs", *Journal of the Association for Information Systems*, Vol. 12 No. 5, pp. 347-374.

Araujo, T. (2018), "Living up to the chatbot hype: the influence of anthropomorphic design cues and communicative agency framing on conversational agent and company perceptions", *Computers in Human Behavior*, Vol. 85, pp. 183-189.

Arnold, V., Clark, N., Collier, P.A., Leech, S.A. and Sutton, S.G. (2006), "The differential use and effect of knowledge-based system explanations in novice and expert judgment decisions", *MIS Quarterly*, Vol. 30 No. 1, pp. 79-97.

Baron, R.M. and Kenny, D.A. (1986), "The moderator–mediator variable distinction in social psychological research: conceptual, strategic, and statistical considerations", *Journal of Personality and Social Psychology*, Vol. 51 No. 6, pp. 1173-1182.

Beldad, A., Hegner, S. and Hoppen, J. (2016), "The effect of virtual sales agent (VSA) gender–product gender congruence on product advice credibility, trust in VSA and online vendor, and purchase intention", *Computers in Human Behavior*, Vol. 60, pp. 62-72.

Brandtzaeg, P.B. and Følstad, A. (2018), "Chatbots: changing user needs and motivations", *Interactions*, Vol. 25 No. 5, pp. 38-43.

Campbell, D.J. (1988), "Task complexity: a review and analysis", *Academy of Management Review*, Vol. 13 No. 1, pp. 40-52.

Chan, H.C., Goswami, S. and Kim, H.-W. (2012), "An alternative fit through problem representation in cognitive fit theory", *Journal of Database Management*, Vol. 23 No. 2, pp. 22-43.

Chan, S.H., Song, Q., Sarker, S. and Plumlee, R.D. (2017), "Decision support system (DSS) use and decision performance: DSS motivation and its antecedents", *Information and Management*, Vol. 54 No. 7, pp. 934-947.

Chang, H.H. and Chuang, S.-S. (2011), "Social capital and individual motivations on knowledge sharing: participant involvement as a moderator", *Information and Management*, Vol. 48 No. 1, pp. 9-18.

Chattaraman, V., Kwon, W.-S., Gilbert, J.E. and Ross, K. (2019), "Should AI-based, conversational digital assistants employ social-or task-oriented interaction style? A task-competency and reciprocity perspective for older adults", *Computers in Human Behavior*, Vol. 90, pp. 315-330.

Chu, P. and Spires, E.E. (2003), "Perceptions of accuracy and effort of decision strategies", *Organizational Behavior and Human Decision Processes*, Vol. 91 No. 2, pp. 203-214.

Chung, M., Ko, E., Joung, H. and Kim, S.J. (2020), "Chatbot e-service and customer satisfaction regarding luxury brands", *Journal of Business Research*, Vol. 117, pp. 587-595.

Cooper, R.B. and Haines, R. (2008), "The Influence of workspace awareness on group intellective decision effectiveness", *European Journal of Information Systems*, Vol. 17 No. 6, pp. 631-648.

Dabholkar, P.A., van Dolen, W.M. and de Ruyter, K. (2009), "A dual-sequence framework for B2C relationship formation: moderating effects of employee communication style in online group chat", *Psychology and Marketing*, Vol. 26 No. 2, pp. 145-174.

Dale, R. (2016), "The return of the chatbots", *Natural Language Engineering*, Vol. 22 No. 5, pp. 811-817.

Davern, M., Shaft, T. and Te'eni, D. (2012), "Cognition matters: enduring questions in cognitive IS research", *Journal of the Association for Information Systems*, Vol. 13 No. 4, pp. 273-314.

Drury-Grogan, M.L., Conboy, K. and Acton, T. (2017), "Examining decision characteristics and challenges for agile software development", *Journal of Systems and Software*, Vol. 131, pp. 248-265.

Edwards, C., Edwards, A., Spence, P.R. and Shelton, A.K. (2014), "Is that a bot running the social media feed? Testing the differences in perceptions of communication quality for a human agent and a bot agent on Twitter", *Computers in Human Behavior*, Vol. 33, pp. 372-376.

Feine, J., Gnewuch, U., Morana, S. and Maedche, A. (2019a), "A taxonomy of social cues for conversational agents", *International Journal of Human-Computer Studies*, Vol. 132, pp. 138-161.

Feine, J., Morana, S. and Maedche, A. (2019b), "Leveraging machine-executable descriptive knowledge in design science research: the case of designing socially-adaptive Chatbots", *International Conference on Design Science Research in Information Systems and Technology*, Worcester, Massachusetts, pp. 76-91.

Fryer, L.K., Ainley, M., Thompson, A., Gibson, A. and Sherlock, Z. (2017), "Stimulating and sustaining interest in a language course: an experimental comparison of chatbot and human task partners", *Computers in Human Behavior*, Vol. 75, pp. 461-468.

Garaus, M., Wagner, U. and Kummer, C. (2015), "Cognitive fit, retail shopper confusion, and shopping value: empirical investigation", *Journal of Business Research*, Vol. 68 No. 5, pp. 1003-1011.

Go, E. and Sundar, S.S. (2019), "Humanizing chatbots: the effects of visual, identity and conversational cues on humanness perceptions", *Computers in Human Behavior*, Vol. 97, pp. 304-316.

Goodhue, D.L. and Thompson, R.L. (1995), "Task-technology fit and individual performance", *MIS Quarterly*, Vol. 19 No. 2, pp. 213-236.

Gulz, A. and Haake, M. (2006), "Design of animated pedagogical agents—a look at their look", *International Journal of Human-Computer Studies*, Vol. 64 No. 4, pp. 322-339.

Gupta, A., Li, H. and Sharda, R. (2013), "Should I send this message? Understanding the impact of interruptions, social hierarchy and perceived task complexity on user performance and perceived workload", *Decision Support Systems*, Vol. 55 No. 1, pp. 135-145.

Häubl, G. and Murray, K.B. (2006), "Double agents: assessing the role of electronic product recommendation systems", *Sloan Management Review*, Vol. 47 No. 3, pp. 8-12.

Hair, J., Anderson, R., Tatham, R. and Black, W. (2010), *Multivariate Data Analysis*, Prentice-Hall, New Jersey.

Hardin, A., Looney, C.A. and Moody, G.D. (2017), "Assessing the credibility of decisional guidance delivered by information systems", *Journal of Management Information Systems*, Vol. 34 No. 4, pp. 1143-1168.

Hill, J., Ford, W.R. and Farreras, I.G. (2015), "Real conversations with artificial intelligence: a comparison between human–human online conversations and human–chatbot conversations", *Computers in Human Behavior*, Vol. 49, pp. 245-250.

Hirsch, B., Seubert, A. and Sohn, M. (2015), "Visualisation of data in management accounting reports: how supplementary graphs improve every-day management judgments", *Journal of Applied Accounting Research*, Vol. 16 No. 2, pp. 221-239.

Ho, A., Hancock, J. and Miner, A.S. (2018), "Psychological, relational, and emotional effects of self-disclosure after conversations with a chatbot", *Journal of Communication*, Vol. 68 No. 4, pp. 712-733.

Holtgraves, T., Ross, S.J., Weywadt, C. and Han, T. (2007), "Perceiving artificial social agents", *Computers in Human Behavior*, Vol. 23 No. 5, pp. 2163-2174.

Holzwarth, M., Janiszewski, C. and Neumann, M.M. (2006), "The influence of avatars on online consumer shopping behavior", *Journal of Marketing*, Vol. 70 No. 4, pp. 19-36.

Hong, W., Thong, J.Y. and Tam, K.Y. (2004), "The effects of information format and shopping task on consumers' online shopping behavior: a cognitive fit perspective", *Journal of Management Information Systems*, Vol. 21 No. 3, pp. 149-184.

Hong, W., Thong, J.Y. and Tam, K.Y. (2007), "How do Web users respond to non-banner-ads animation? The effects of task type and user experience", *Journal of the American Society for Information Science and Technology*, Vol. 58 No. 10, pp. 1467-1482.

Hubspot (2018), "Hubspot research consumer customer support survey", available at: https://blog.hubspot.com/news-trends/live-chat-go-to-market-flaw (accessed 10 May 2020).

Janiszewski, C. (1998), "The influence of display characteristics on visual exploratory search behavior", *Journal of Consumer Research*, Vol. 25 No. 3, pp. 290-301.

Jiang, Z. and Benbasat, I. (2007), "The effects of presentation formats and task complexity on online consumers' product understanding", *MIS Quarterly*, Vol. 31 No. 3, pp. 475-500.

John, T. and Kundisch, D. (2015), "Why fit leads to surprise: an extension of cognitive fit theory to creative problems", *International Conference on Information Systems*, Texas, USA, pp. 1-19.

Kamis, A., Koufaris, M. and Stern, T. (2008), "Using an attribute-based decision support system for user-customized products online: an experimental investigation", *MIS Quarterly*, Vol. 32 No. 1, pp. 159-177.

Keeling, K., McGoldrick, P. and Beatty, S. (2010), "Avatars as salespeople: communication style, trust, and intentions", *Journal of Business Research*, Vol. 63 No. 8, pp. 793-800.

Kelton, A.S., Pennington, R.R. and Tuttle, B.M. (2010), "The effects of information presentation format on judgment and decision making: a review of the information systems research", *Journal of Information Systems*, Vol. 24 No. 2, pp. 79-105.

Khatri, V., Vessey, I., Ram, S. and Ramesh, V. (2006), "Cognitive fit between conceptual schemas and internal problem representations: the case of geospatio-temporal conceptual schema

comprehension", *IEEE Transactions on Professional Communication*, Vol. 49 No. 2, pp. 109-127.

Kim, I., Jeon, S.M. and Hyun, S.S. (2011), "The role of effective service provider communication style in the formation of restaurant patrons' perceived relational benefits and loyalty", *Journal of Travel and Tourism Marketing*, Vol. 28 No. 7, pp. 765-786.

Köhler, C.F., Rohm, A.J., de Ruyter, K. and Wetzels, M. (2011), "Return on interactivity: the impact of online agents on newcomer adjustment", *Journal of Marketing*, Vol. 75 No. 2, pp. 93-108.

Komiak, S.Y. and Benbasat, I. (2006), "The effects of personalization and familiarity on trust and adoption of recommendation agents", *MIS Quarterly*, Vol. 30 No. 4, pp. 941-960.

Komiak, S.Y., Wang, W. and Benbasat, I. (2004), "Trust building in virtual salespersons versus in human salespersons: similarities and differences", *E-Service*, Vol. 3 No. 3, pp. 49-64.

Kumar, N. and Benbasat, I. (2004), "The effect of relationship encoding, task type, and complexity on information representation: an empirical evaluation of 2D and 3D line graphs", *MIS Quarterly*, Vol. 28 No. 2, pp. 255-281.

Kumar, N. and Benbasat, I. (2006), "Research note: the influence of recommendations and consumer reviews on evaluations of websites", *Information Systems Research*, Vol. 17 No. 4, pp. 425-439.

Lee, Y.E. and Benbasat, I. (2010), "Interaction design for mobile product recommendation agents: supporting users' decisions in retail stores", *ACM Transactions on Computer-Human Interaction*, Vol. 17 No. 4, pp. 1-32.

Lee, Y.E. and Benbasat, I. (2011), "The influence of trade-off difficulty caused by preference elicitation methods on user acceptance of recommendation agents across loss and gain conditions", *Information Systems Research*, Vol. 22 No. 4, pp. 867-884.

Lee, H.-K., Suh, K.-S. and Benbasat, I. (2001), "Effects of task-modality fit on user performance", *Decision Support Systems*, Vol. 32 No. 1, pp. 27-40.

Lu, H.-P. and Yang, Y.-W. (2014), "Toward an understanding of the behavioral intention to use a social networking site: an extension of task-technology fit to social-technology fit", *Computers in Human Behavior*, Vol. 34, pp. 323-332.

Luo, X., Tong, S., Fang, Z. and Qu, Z. (2019), "Frontiers: machines vs. humans: the impact of artificial intelligence chatbot disclosure on customer purchases", *Marketing Science*, Vol. 38 No. 6, pp. 937-947.

Luszczynska, A., Scholz, U. and Schwarzer, R. (2005), "The general self-efficacy scale: multicultural validation studies", *Journal of Psychology*, Vol. 139 No. 5, pp. 439-457.

Malinowski, B. (1994), "The problem of meaning in primitive languages", in Maybin, J. (Ed.), *Language and Literacy in Social Practice: A Reader*, Multingual Matters, Clevedon, pp. 1-10.

Marchionini, G. and Shneiderman, B. (1993), "Finding facts vs. browsing knowledge in hypertext systems", in Shneiderman, B. (Ed.), *Sparks of Innovation in Human-Computer Interaction*, Ablex, Norwood, NJ, pp. 99-119.

McDonald, D.M. and Chen, H. (2006), "Summary in context: searching versus browsing", *ACM Transactions on Information Systems*, Vol. 24 No. 1, pp. 111-141.

McNamara, K., Dennis, A.R. and Carte, T.A. (2008), "It's the thought that counts: the mediating effects of information processing in virtual team decision making", *Information Systems Management*, Vol. 25 No. 1, pp. 20-32.

McTear, M., Callejas, Z. and Griol, D. (2016), *The Conversational Interface: Talking to Smart Devices*, Springer.

Melis, K., Campo, K., Breugelmans, E. and Lamey, L. (2015), "The impact of the multi-channel retail mix on online store choice: does online experience matter?", *Journal of Retailing*, Vol. 91 No. 2, pp. 272-288.

Mertins, L. and White, L.F. (2016), "Presentation formats, performance outcomes, and implications for performance evaluations", in Epstein, M.J. and Malina, M.A. (Eds), *Advances in Management Accounting*, Emerald Group Publishing, pp. 1-34.

Mimoun, M.S.B., Poncin, I. and Garnier, M. (2017), "Animated conversational agents and e-consumer productivity: the roles of agents and individual characteristics", *Information and Management*, Vol. 54 No. 5, pp. 545-559.

Morana, S., Schacht, S., Scherp, A. and Maedche, A. (2014a), *Conceptualization and Typology of Guidance in Information Systems*, working paper in Information Systems Series, University of Mannheim, Mannheim, Germany, 14 August.

Morana, S., Schacht, S., Scherp, A. and Maedche, A. (2014b), "Designing a process guidance system to support user's business process compliance", *International Conference on Information Systems*, Auckland, New Zealand, pp. 1-19.

Mukaka, M.M. (2012), "A guide to appropriate use of correlation coefficient in medical research", *Malawi Medical Journal*, Vol. 24 No. 3, pp. 69-71.

Olston, C. and Chi, E.H. (2003), "ScentTrails: integrating browsing and searching on the Web", *ACM Transactions on Computer-Human Interaction*, Vol. 10 No. 3, pp. 177-197.

Oracle (2016), "Can virtual experiences replace reality? The future role for humans in delivering customer experience", available at: https://www.oracle.com/webfolder/s/delivery_production/docs/FY16h1/doc35/CXResearchVirtualExperiences.pdf (accessed 10 May 2020).

Parkes, A. (2013), "Persuasive decision support: improving reliance on decision aids", *Pacific Asia Journal of the Association for Information Systems*, Vol. 4 No. 3, pp. 1-13.

Pavlou, P.A. and Fygenson, M. (2006), "Understanding and predicting electronic commerce adoption: an extension of the theory of planned behavior", *MIS Quarterly*, Vol. 30 No. 1, pp. 115-143.

Payne, J.W., Bettman, J.R. and Johnson, E.J. (1990), "The adaptive decision maker: effort and accuracy in choice", in Hogarth, R.M. (Ed.), *Insights in Decision Making: A Tribute to Hillel J. Einhorn*, University of Chicago Press, Chicago, IL, pp. 129-153.

Perdana, A., Robb, A. and Rohde, F. (2018), "Interactive data visualization for accounting information: a three-fit perspective", *Behaviour and Information Technology*, Vol. 38 No. 1, pp. 1-16.

Poncin, I., Mimoun, B., Slim, M. and Garnier, M. (2013), "Improving consumer performance and perceived service quality in a new type of commercial environment in 3D, thanks to embodied virtual sales agents", in El-Gohary, H. and Eid, R. (Eds), *E-marketing in Developed and Developing Countries: Emerging Practices*, IGI Global, pp. 233-245.

Qiu, L. and Benbasat, I. (2009), "Evaluating anthropomorphic product recommendation agents: a social relationship perspective to designing information systems", *Journal of Management Information Systems*, Vol. 25 No. 4, pp. 145-182.

Saad, S.B. and Abida, F.C. (2016), "Social interactivity and its impact on a user's approach behavior in commercial web sites: a study case of virtual agent presence", *Journal of Marketing Management*, Vol. 4 No. 2, pp. 63-80.

Sanchez, O., Costa, P.H.S.E. and Goes, P. (2015), "Shaping customer confidence in online purchasing decision: the role of DSS tools supporting an information aggregator", *Hawaii International Conference on System Sciences*, Kauai, HI, USA, pp. 4294-4303.

Schwarz, A. and Schwarz, C. (2007), "The role of latent beliefs and group cohesion in predicting group decision support systems success", *Small Group Research*, Vol. 38 No. 1, pp. 195-229.

Shaft, T.M. and Vessey, I. (2006), "The role of cognitive fit in the relationship between software comprehension and modification", *MIS Quarterly*, Vol. 30 No. 1, pp. 29-55.

Silver, M.S. (1991), "Decisional guidance for computer-based decision support", *MIS Quarterly*, Vol. 15 No. 1, pp. 105-122.

Silver, M. (2006), "Decisional guidance: broadening the scope", in Zhang, P. and Galletta, D. (Eds), *Human–Computer Interaction and Management Information Systems: Foundations*, M. E. Sharpe, Inc., Armonk, New York, pp. 90-119.

Speier, C. (2006), "The influence of information presentation formats on complex task decision-making performance", *International Journal of Human-Computer Studies*, Vol. 64 No. 11, pp. 1115-1131.

Speier, C. and Morris, M.G. (2003), "The influence of query interface design on decision-making performance", *MIS Quarterly*, Vol. 27 No. 3, pp. 397-423.

Sweller, J. (1988), "Cognitive load during problem solving: effects on learning", *Cognitive Science*, Vol. 12 No. 2, pp. 257-285.

Todd, P. and Benbasat, I. (1991), "An experimental investigation of the impact of computer based decision aids on decision making strategies", *Information Systems Research*, Vol. 2 No. 2, pp. 87-115.

Van den Broeck, E., Zarouali, B. and Poels, K. (2019), "Chatbot advertising effectiveness: when does the message get through?", *Computers in Human Behavior*, Vol. 98, pp. 150-157.

Van Dolen, W.M., Dabholkar, P.A. and De Ruyter, K. (2007), "Satisfaction with online commercial group chat: the influence of perceived technology attributes, chat group characteristics, and advisor communication style", *Journal of Retailing*, Vol. 83 No. 3, pp. 339-358.

Verhagen, T., Van Nes, J., Feldberg, F. and Van Dolen, W. (2014), "Virtual customer service agents: using social presence and personalization to shape online service encounters", *Journal of Computer-Mediated Communication*, Vol. 19 No. 3, pp. 529-545.

Vessey, I. (1991), "Cognitive fit: a theory-based analysis of the graphs versus tables literature", *Decision Sciences*, Vol. 22 No. 2, pp. 219-240.

Vessey, I. and Galletta, D. (1991), "Cognitive fit: an empirical study of information acquisition", *Information Systems Research*, Vol. 2 No. 1, pp. 63-84.

Wang, W. and Benbasat, I. (2007), "Recommendation agents for electronic commerce: effects of explanation facilities on trusting beliefs", *Journal of Management Information Systems*, Vol. 23 No. 4, pp. 217-246.

Wang, W. and Benbasat, I. (2009), "Interactive decision aids for consumer decision making in e-commerce: the influence of perceived strategy restrictiveness", *MIS Quarterly*, Vol. 33 No. 2, pp. 293-320.

Wang, W. and Benbasat, I. (2012), "Research note—a contingency approach to investigating the effects of user - system interaction modes of online decision aids", *Information Systems Research*, Vol. 24 No. 3, pp. 861-876.

Wang, W. and Benbasat, I. (2016), "Empirical assessment of alternative designs for enhancing different types of trusting beliefs in online recommendation agents", *Journal of Management Information Systems*, Vol. 33 No. 3, pp. 744-775.

Wang, L.C., Baker, J., Wagner, J.A. and Wakefield, K. (2007), "Can a retail web site be social?", *Journal of Marketing*, Vol. 71 No. 3, pp. 143-157.

Weizenbaum, J. (1966), "ELIZA—a computer program for the study of natural language communication between man and machine", *Communications of the ACM*, Vol. 9 No. 1, pp. 36-45.

White, R.W. and Roth, R.A. (2009), "Exploratory search: beyond the query-response paradigm", *Synthesis Lectures on Information Concepts, Retrieval, and Services*, Vol. 1 No. 1, pp. 1-98.

Wood, N.T., Solomon, M.R. and Englis, B.G. (2005), "Personalisation of online avatars: is the messenger as important as the message?", *International Journal of Internet Marketing and Advertising*, Vol. 2 Nos 1-2, pp. 143-161.

Xiao, B. and Benbasat, I. (2007), "E-commerce product recommendation agents: use, characteristics, and impact", *MIS Quarterly*, Vol. 31 No. 1, pp. 137-209.

Xiao, B. and Benbasat, I. (2014), "Research on the use, characteristics, and impact of e-commerce product recommendation agents: a review and update for 2007–2012", in Martínez-López, F.J. (Ed.), *Handbook of Strategic E-Business Management*, Springer, Berlin, Heidelberg, pp. 403-431.

Xiao, B. and Benbasat, I. (2018), "An empirical examination of the influence of biased personalized product recommendations on consumers' decision making outcomes", *Decision Support Systems*, Vol. 110, pp. 46-57.

Xu, J.D. (2016), "Retaining customers by utilizing technology-facilitated chat: mitigating website anxiety and task complexity", *Information and Management*, Vol. 53 No. 5, pp. 554-569.

Xu, J., Benbasat, I. and Cenfetelli, R.T. (2014a), "The influences of online service technologies and task complexity on efficiency and personalization", *Information Systems Research*, Vol. 25 No. 2, pp. 420-436.

Xu, J.D., Benbasat, I. and Cenfetelli, R.T. (2014b), "The nature and consequences of trade-off transparency in the context of recommendation agents", *MIS Quarterly*, Vol. 38 No. 2, pp. 379-406.

Yan, D. and Chen, L. (2015), "An empirical study of user decision making behavior in e-commerce", *International Conference on HCI in Business*, Los Angeles, USA, pp. 414-426.

Yang, J., Zheng, R., Zhao, L. and Gupta, S. (2017), "Enhancing customer brand experience and loyalty through enterprise microblogs: empirical evidence from a communication framework perspective", *Information Technology and People*, Vol. 30 No. 3, pp. 580-601.

Yin, M. and Zhai, S. (2006), "The benefits of augmenting telephone voice menu navigation with visual browsing and search", *SIGCHI Conference on Human Factors in Computing Systems*, Montreal, Canada, pp. 319-328.

Yoon, V.Y., Hostler, R.E., Guo, Z. and Guimaraes, T. (2013), "Assessing the moderating effect of consumer product knowledge and online shopping experience on using recommendation agents for customer loyalty", *Decision Support Systems*, Vol. 55 No. 4, pp. 883-893.

Yorkston, E. and Menon, G. (2004), "A sound idea: phonetic effects of brand names on consumer judgments", *Journal of Consumer Research*, Vol. 31 No. 1, pp. 43-51.

Zha, X., Li, J. and Yan, Y. (2013), "Information self-efficacy and information channels: decision quality and online shopping satisfaction", *Online Information Review*, Vol. 37 No. 6, pp. 872-890.

Zhu, D.H., Chang, Y.P., Luo, J.J. and Li, X. (2014), "Understanding the adoption of location-based recommendation agents among active users of social networking sites", *Information Processing and Management*, Vol. 50 No. 5, pp. 675-682.

Appendix

The appendix is available online for this article.

About the authors

Jengchung Victor Chen is a distinguished professor at Institute of International Management, College of Management, National Cheng Kung University, Taiwan. He obtained his PhD in CIS from University of Hawaii. His research focuses on Electronic Commerce, Information Ethics, Project Management, IS Service Quality and has been published in journals such as Decision Support Systems, Journal of the Association for Information Systems, Information and Management, International Journal of Electronic Commerce, European Journal of Information Systems and other outlets. Jengchung Victor Chen is the corresponding author and can be contacted at: victor@mail.ncku.edu.tw

Huyen Thi Le is an MBA holder and graduated from Institute of International Management, College of Management, National Cheng Kung University, Taiwan. Her research focuses on Decision Support Systems, Electronic Commerce and Social Media.

Sinh Thi Thu Tran is PhD candidate at Institute of International Management, College of Management, National Cheng Kung University, Taiwan. Her research interests include Electronic Commerce, Social Media, Information Ethics and Project Management and have been published in Industrial Management and Data Systems.

For instructions on how to order reprints of this article, please visit our website:
www.emeraldgrouppublishing.com/licensing/reprints.htm
Or contact us for further details: **permissions@emeraldinsight.com**

Reproduced with permission of copyright owner. Further reproduction prohibited without permission.